

Exploring Pascal's Triangle

September 13th, 2026

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Warm-Up

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- For the first spot there are 5 possible students who can stand in that spot. Select one student and place them there.
- Now, the second spot has 4 possible students. Select one student and place them there.
- Now, the third spot has 3 possible students. Select one student and place them there.
- Now, the fourth spot has 2 possible students. Select one student and place them there.
- The fifth spot has 1 possible student. That student goes to that spot.

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- The fifth spot has 1 possible student. That student goes to that spot.

In total, this is $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ ways.

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For example, $0! = 1$, $1! = 1$, $2! = 2$, $3! = 6$, $4! = 24$, and $5! = 120$.

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How many ways are there to select a president, a vice president, and a secretary from a group of 8 people?

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How many ways are there to select a president, a vice president, and a secretary from a group of 8 people?

- There are 8 ways to select the president.
- Among the remaining people, there are 7 ways to select the vice president.
- Among the remaining people, there are 6 ways to select the secretary.

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- There are 8 ways to select the president.
- Among the remaining people, there are 7 ways to select the vice president.
- Among the remaining people, there are 6 ways to select the secretary.

In total, there are $8 \cdot 7 \cdot 6$ ways to select the group.

Warm-Up (Round 3)

Problem

How many ways are there to select a group of three people from a group of 8 people?

Recall, there are $8 \cdot 7 \cdot 6$ ways to select a president, vice president, and secretary. In that case, the *order* of the way we select the people matter.

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$$\frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = \frac{8!}{5!3!} = 56.$$

“Choose”

The phrase “ n choose k ” is denoted by $\binom{n}{k}$ and can be expressed as

$$\binom{n}{k} = \frac{n!}{(n-k)!k!},$$

the number of ways to select k objects from n objects where order does not matter.

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- $$\binom{n}{k} = \binom{n}{n-k},$$

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- $\binom{n-1}{k-1} + \binom{n-1}{k} = \binom{n}{k},$ and
- $\binom{k}{k} + \binom{k+1}{k} + \cdots + \binom{n}{k} = \binom{n+1}{k+1}$

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Simplify $(1 + x)^n$.

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We want to compute coefficient of x^k for $0 \leq k \leq n$:

$$(1 + x)^n = \underbrace{(1 + x)(1 + x)(1 + x) \cdots (1 + x)(1 + x)}_{n \text{ occurrences}}.$$

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- Suppose we expand *without* simplifying. To get each term, in each $(1 + x)$, we either multiply by 1 or x .
- Of the n occurrences of $(1 + x)$, we must select k of the occurrences to be x and $n - k$ of the occurrences to be 1.
- The number of ways to select k occurrences to be x is $\binom{n}{k}$, so, the coefficient of x^k must be $\binom{n}{k}$.

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As the coefficient of x^k is $\binom{n}{k}$, then $(1 + x)^n$ can be expressed as

$$\binom{n}{0}x^0 + \binom{n}{1}x^1 + \binom{n}{2}x^2 + \cdots + \binom{n}{n-1}x^{n-1} + \binom{n}{n}x^n.$$

The Triangle

Suppose we place all the entries into a nice triangle!

$$\begin{array}{ccccccccccc} & & & & \binom{0}{0} & & & & & & \\ & & & \binom{1}{0} & & \binom{1}{1} & & & & & \\ & & \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2} & & & & \\ & \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} & & & \\ & \binom{4}{0} & & \binom{4}{1} & & \binom{4}{2} & & \binom{4}{3} & & \binom{4}{4} & \\ \binom{5}{0} & & \binom{5}{1} & & \binom{5}{2} & & \binom{5}{3} & & \binom{5}{4} & & \binom{5}{5} \\ \binom{6}{0} & \binom{6}{1} & \binom{6}{2} & \binom{6}{3} & \binom{6}{4} & \binom{6}{5} & \binom{6}{6} \end{array}$$

The Triangle

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				0								
			1		1							
		1		2		1						
	1		3		3		1					
	1		4		6		4		1			
	1		5		10		10		5		1	
1		6		15		20		15		6		1

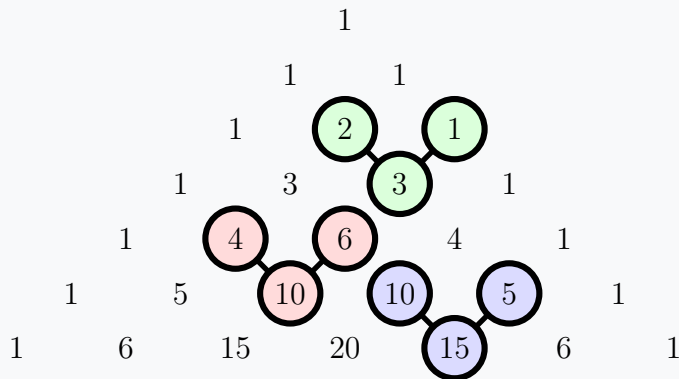
The Triangle

Each row has the coefficients to $(x + 1)^n$.

$$\begin{array}{ccccccccccc} & & & & & & & & 1 & & & & & & & & & \\ & & & & & & & & x & & 1 & & & & & & & & \\ & & & & & & & x^2 & & 2x & & 1 & & & & & & & \\ & & & x^3 & & 3x^2 & & 3x & & 1 & & & & & & & & & \\ & & x^4 & & 4x^3 & & 6x^2 & & 4x & & 1 & & & & & & & & \\ & x^5 & & 5x^4 & & 10x^3 & & 10x^2 & & 5x & & 1 & & & & & & & \\ x^6 & & 6x^5 & & 15x^4 & & 20x^3 & & 15x^2 & & 6x & & 1 & & & & & & \end{array}$$

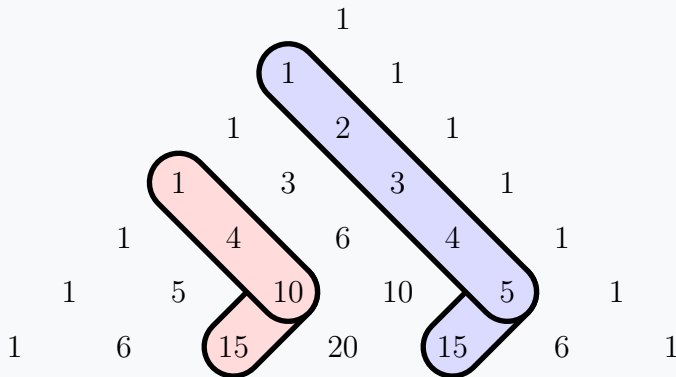
The Triangle

We can see $\binom{n-1}{k-1} + \binom{n-1}{k} = \binom{n}{k}$.



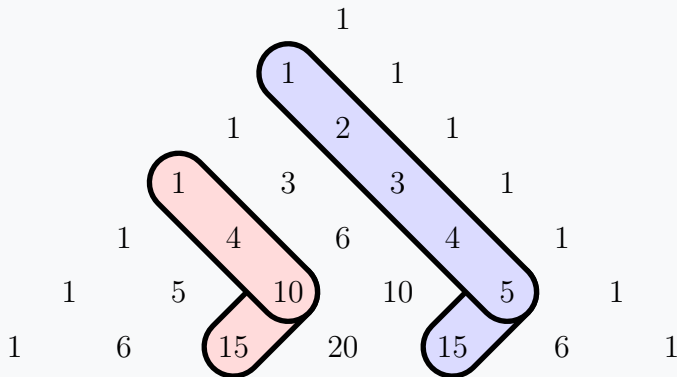
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Overall, the Pascal's Triangle gives us a really nice visual understanding of the choose function.

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Definition

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Theorem (Legendre's Theorem)

For a positive integer n , then

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How do we prove this?

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Theorem (Legendre's Theorem)

For a positive integer n and prime p , then

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Summing implies $v_p(n!) = \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor$.

Base Representation

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$$\begin{aligned} 2026_{10} &= 1 \cdot 2^{10} + 1 \cdot 2^9 + 1 \cdot 2^8 + 1 \cdot 2^7 + 1 \cdot 2^6 + 1 \cdot 2^5 \\ &\quad + 0 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 \\ &= 11111101010_2 \end{aligned}$$

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The major rule is that in a base- b representation of a base-10 number n requires that all digits are between 0 and $b - 1$, inclusive. For example, all digits in a base-2 representation are either 0 or 1.

Base Representation

Definition

For a positive integer p , let the base- p representation of n be

$$(a_p a_{p-1} \cdots a_1 a_0)_p.$$

Then, we define

$$S_p(n) = a_p + a_{p-1} + \cdots + a_1 + a_0$$

to be the sum of the digits of the base- p representation of n .

Legendre's Theorem (Part 2.)

Theorem (Legendre's Theorem)

For a positive integer n and prime p , then

$$v_p(n!) = \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor = \frac{n - S_p(n)}{p - 1}.$$

Let the base p representation of n be

$$(e_x e_{x-1} \cdots e_1 e_0)_p$$

where the e_i are digits in base p .

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where the e_i are digits in base p . Notice,

$$e_x \cdot p^x + e_{x-1} \cdot p^{x-1} + \cdots + e_i \cdot p^i + \left[e_{i-1} \cdot p^{i-1} + e_1 \cdot p^1 + e_0 \cdot p^0 \right].$$

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As $[e_{i-1} \cdot p^{i-1} + e_1 \cdot p^1 + e_0 \cdot p^0] < p^i$, the base- p representation is

$$\left\lfloor \frac{n}{p^i} \right\rfloor = (e_x e_{x-1} \cdots e_i)_p.$$

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Theorem (Legendre's Theorem)

For a positive integer n and prime p , then

$$\nu_p(n!) = \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor = \frac{n - S_p(n)}{p - 1}.$$

Notice,

$$\begin{aligned} \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor &= \sum_{j=1}^x e_j \cdot (p^{j-1} + p^{j-2} + \cdots + 1) = \sum_{j=1}^x e_j \left(\frac{p^j - 1}{p - 1} \right) \\ &= \frac{\sum_{j=1}^x e_j p^j - \sum_{j=1}^x e_j}{p - 1} = \frac{(n - e_0) - (S_p(n) - e_0)}{p - 1} = \frac{n - S_p(n)}{p - 1}. \end{aligned}$$

Kummer's Theorem

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For positive integers m and n , and prime p , then

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Notice, by Legendre's

$$\begin{aligned} & \nu_p(n!) - \nu_p((n - m)!) - \nu_p(m!) \\ &= \frac{n - S_p(n)}{p - 1} - \frac{(n - m) - S_p(n - m)}{p - 1} - \frac{m - S_p(m)}{p - 1} \\ &= \frac{S_p(m) + S_p(n - m) - S_p(n)}{p - 1}. \end{aligned}$$

The Big Question

Problem

Let m and n be non-negative integers. When is $\binom{n}{m}$ even? When is $\binom{n}{m}$ odd?

Note that

$$\nu_2 \left(\binom{n}{m} \right) = S_2(m) + S_2(n - m) - S_2(n).$$

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Consider

$$\begin{array}{cccc}
 & 1 & 0 & 0 & 1 \\
 + & & 1 & 0 & 1 \\
 \hline
 & 1 & 1 & \text{red } 0 \rightarrow 1 & \text{red } 2 \rightarrow 0
 \end{array}$$

where m is 1001_2 and $n - m = 101_2$.

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where m is 1001_2 and $n - m = 101_2$. Each time we make a carry the result in one column decreases by 2 and increase another by 1. So, $S_2(m) + S_2(n - m) - S_2(n)$ really is just the number of carries.

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Note that $\binom{n}{m}$ is odd when

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and so no such carries exist when adding m and $n - m$.

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Note that $\binom{n}{m}$ is odd when

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and so no such carries exist when adding m and $n - m$. This means for the value to be odd, then each digit in the binary representation of m is at most the corresponding digit in the binary representation of n .

Neat Corollary

For a positive integer n , the number of positive integers m for which the binomial $\binom{n}{m}$ is an odd integer is

$$2^{S_2(n)}.$$

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For a positive integer n , the number of positive integers m for which the binomial $\binom{n}{m}$ is an odd integer is

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This is because suppose

$$n = (a_p a_{p-1} \cdots a_1 a_0)_2.$$

If $a_k = 1$, then m has two options for the corresponding digit.

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For a positive integer n , the number of positive integers m for which the binomial $\binom{n}{m}$ is an odd integer is

$$2^{S_2(n)}.$$

This is because suppose

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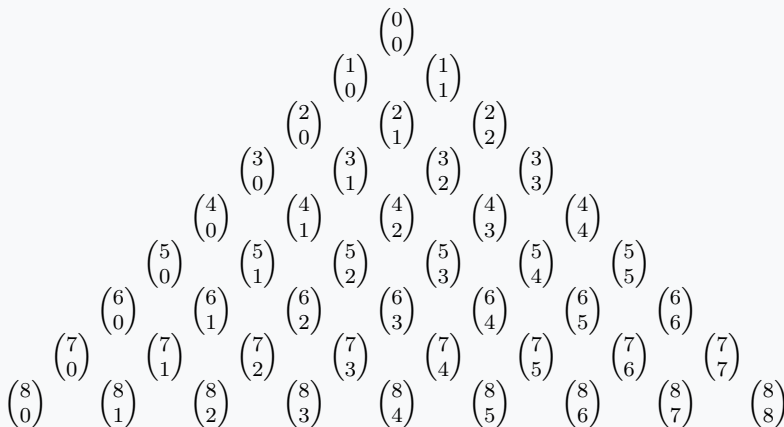
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If $a_k = 1$, then m has two options for the corresponding digit. If $a_k = 0$, then the corresponding digit must be 0. We therefore multiply by 2 for each digit that is 1 in the binary representation of n implying the result.

The Big Question

Let's look back at the Pascal's Triangle:



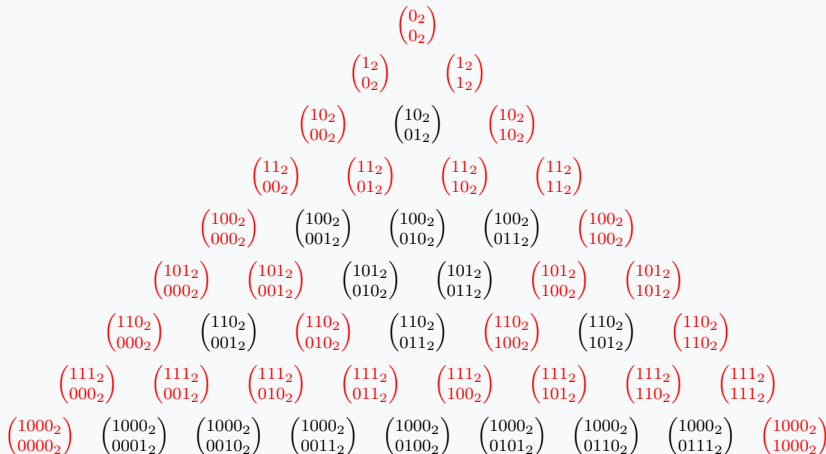
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$$\begin{array}{cccccccccc}
 & & & & & & & & & & \binom{0_2}{0_2} \\
 & & & & & & & & & & & \binom{1_2}{0_2} & \binom{1_2}{1_2} \\
 & & & & & & & & & & \binom{10_2}{00_2} & \binom{10_2}{01_2} & \binom{10_2}{10_2} \\
 & & & & & & & & & \binom{11_2}{00_2} & \binom{11_2}{01_2} & \binom{11_2}{10_2} & \binom{11_2}{11_2} \\
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 \binom{1000_2}{0000_2} & \binom{1000_2}{0001_2} & \binom{1000_2}{0010_2} & \binom{1000_2}{0011_2} & \binom{1000_2}{0100_2} & \binom{1000_2}{0101_2} & \binom{1000_2}{0110_2} & \binom{1000_2}{0111_2} & \binom{1000_2}{1000_2}
 \end{array}$$

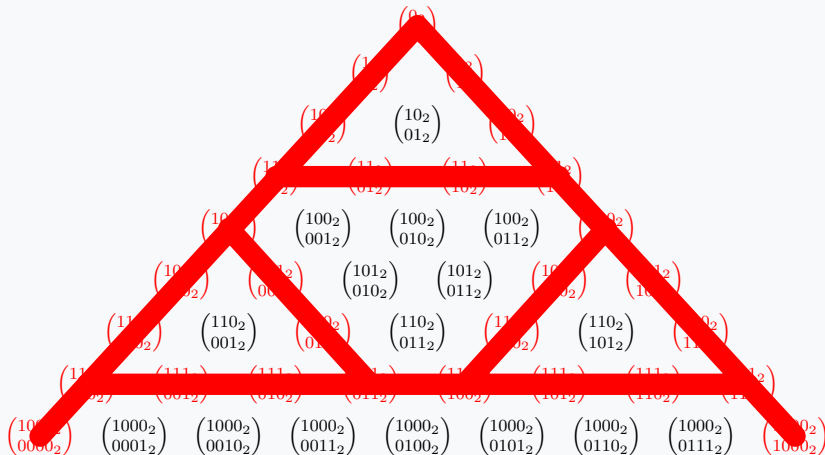
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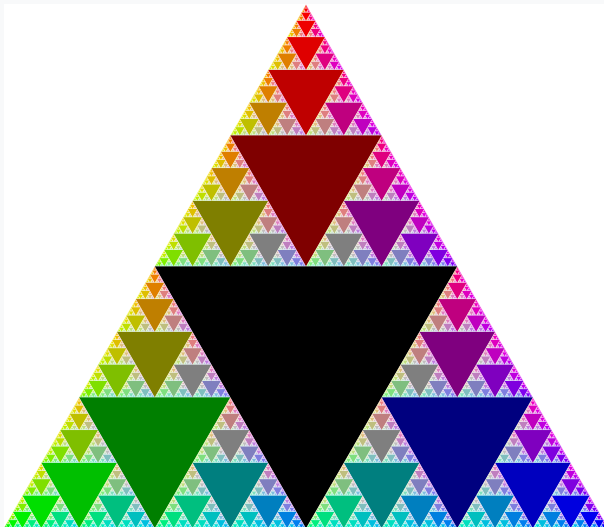
The Big Question

Let's look back at the Pascal's Triangle:



Sierpinski's Triangle

Eventually, it will look like the **Sierpinski's Triangle**!



Let's see why!

Suppose the first 2^n rows; that is, rows 0 to $2^n - 1$ follow the Sierpinski's Triangle. We will show rows 0 to $2^{n+1} - 1$ follow the Sierpinski's Triangle.

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$$0 \leq k = (a_{n-1}a_{n-2} \cdots a_1a_0)_2 < 2^n, \text{ and}$$

$$0 \leq 2^n + k = (1a_{n-1}a_{n-2} \cdots a_1a_0)_2 < 2^{n+1}.$$

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Conclusion: If $\binom{k}{j}$ is odd, then $\binom{2^n+k}{j}$ and $\binom{2^n+k}{2^n+j}$ are both odd.

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This is even to imply the Pascal's Triangle does follow the Sierpinski's Triangle!

Further Explanation

Suppose

$$m = (a_p a_{p-1} \cdots a_1 a_0)_2, \text{ and}$$

$$n = (b_p b_{p-1} \cdots b_1 b_0)_2.$$

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Let's investigate this more! We want to know values of the Pascal's Triangle divisible by 3, 5, and so on!

A New Theorem!

Theorem

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At least n powers of p divide the numerator and at most $n - 1$ powers divide the denominator. Thus, the right-hand side is divisible by p .

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These observations show that in the binomial theorem expansion of $(1 + x)^{p^n}$, only the first ($k = 0$) and last ($k = p^n$) coefficients are non-zero $\pmod{p}$, and our desired result follows.

Let's use this!

Suppose $p = 5$, $r = 482$, and $c = 176$. Note

$$482 = 3 \cdot 5^3 + 4 \cdot 5^2 + 1 \cdot 5 + 2 = 3412_5,$$

$$176 = 1 \cdot 5^3 + 2 \cdot 5^2 + 0 \cdot 5 + 1 = 1201_5.$$

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Using the lemma allows us to write

$$\begin{aligned}(1+x)^{482} &= (1+x)^{3(125)} \cdot (1+x)^{4(25)} \cdot (1+x)^{1(5)} \cdot (1+x)^2 \\ &\equiv (1+x^{125})^3 \cdot (1+x^{25})^4 \cdot (1+x^5)^1 \cdot (1+x)^2 \pmod{5}.\end{aligned}$$

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$$\binom{482}{176} \equiv \binom{3}{1} \binom{4}{2} \binom{1}{0} \binom{2}{1} \pmod{5}.$$

What does this mean?

Theorem (Lucas's Theorem)

Let p be a prime number, and let r and c be written in p -ary notation:

$$r = r_k \cdots r_2 r_1 r_0 = r_0 + r_1 p + r_2 p^2 + \cdots + r_k p^k \quad (0 \leq r_i < p)$$

$$c = c_k \cdots c_2 c_1 c_0 = c_0 + c_1 p + c_2 p^2 + \cdots + c_k p^k \quad (0 \leq c_i < p)$$

Then

$$\binom{r}{c} \equiv \binom{r_0}{c_0} \binom{r_1}{c_1} \binom{r_2}{c_2} \cdots \binom{r_k}{c_k} \pmod{p}.$$

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This is much stronger version of the Kummer's Theorem we saw before! The Kummer's Theorem will tell us if a certain binomial is divisible by a prime p . On the other hand, Lucas's Theorem tells us the actual remainder when divided by p !

Questions?

- What happens to the Pascal's Triangle if you color red the elements which are divisible by 3 and white the elements which are not divisible by 3? How about for other primes like 5 or 7? How about for prime powers like 4, 8, 9, 16, or 27? How about for composites like 6?

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- What happens to the Pascal's Triangle if you color red the elements which $0 \pmod{3}$, blue the elements which are $1 \pmod{3}$, and green the elements which are $2 \pmod{3}$? How about for other primes like 5 or 7? How about for prime powers like 4, 8, 9, 16, or 27? How about for composites like 6?

Problems

Problem

For a positive integer n , show that

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n.$$

Note that

$$(1+x)^n = \binom{n}{0}x^0 + \binom{n}{1}x^1 + \cdots + \binom{n}{n}x^n.$$

Plugging in $x = 1$, then

$$2^n = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n}.$$

Problems

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For a positive integer n , show that

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Consider a set of n elements $\{s_1, s_2, \dots, s_n\}$.

- The number of sets of size 0 is $\binom{n}{0}$,
- The number of sets of size 1 is $\binom{n}{1}$,
- The number of sets of size 2 is $\binom{n}{2}$,
- and so on.

Summing together this should also equal 2^n .

Problems

Problem

Let S be a subset of $\{0, 1, \dots, 1023\}$ such that that for any $a, b \in S$ with $a > b$, the binomial coefficient $\binom{a}{b}$ is even. Determine the greatest possible number of elements in S .

Problems

Problem

How many terms in the expansion

$$(x^2 + x + 1)^{2026}$$

are such that the coefficient is divisible by 3?

Notice $x^2 + x + 1 \equiv x^2 + 4x + 4 = (x + 2)^2 \pmod{3}$. Then,

$$(x^2 + x + 1)^{2026} \equiv (x + 2)^{4052} \equiv (x - 1)^{4052} \pmod{3}.$$

The coefficient of x^k is $\binom{4052}{k} (-1)^{4052-k}$. Notice, $4052_{10} = 12120002_3$.

The number of ways for this to be NOT divisible by 3 is

$$2 \cdot 3 \cdot 2 \cdot 3 \cdot 1 \cdot 1 \cdot 1 \cdot 3 = 108.$$

This comes to be $4053 - 108 = \boxed{3945}$.